

A First Dive into Brownian Motion

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Preface

Cells, the units of life, are sophisticated microscopic machines made of organelles and macromolecules. These components of the cell undergo collisions with molecules of the cytoplasmic fluid (cytosol), leading to erratic motion. Motivated by their significance to us as living beings ourselves, this text is a stepping stone into such dynamics as an exploration of Brownian motion in one dimension. We start our discussion with how position and velocity correlations evolve with time for a Brownian particle and how the ‘Langevin equation’ helps us model its dynamics. We then briefly review some concepts from stochastic calculus and use them to derive the Fokker–Planck equation, a second-order linear PDE describing the evolution of the probability distribution of the position and velocity of such a particle. Finally, we test some results derived in the text via numerical simulation. I am grateful to Physics LibreTexts, other online resources, and the book *Soft Matter Physics* by Masao Doi for helping me write this report.

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1 Brownian Motion and Temporal Correlations

We shall follow the following notational conventions in this section:

- $\langle \cdot \rangle$ denotes ensemble average
- x_i : physical quantity
- $\Delta x_i(t) = x_i(t) - \bar{x}_i$, where $\bar{x}_i = \langle x_i(t) \rangle$ (independent of t)

1.1 Correlation Functions

$$\langle \Delta x_i(t_1) \Delta x_j(t_2) \rangle$$

- $i = j$: autocorrelation
- $i \neq j$: cross-correlation

By time translational invariance (equilibrium):

$$\langle \Delta x_i(t_1 - \tau) \Delta x_j(t_2 - \tau) \rangle = \langle \Delta x_i(t_1) \Delta x_j(t_2) \rangle$$

Thus correlation depends only on $\tau = t_2 - t_1$. This also gives:

$$\langle \Delta x_i(t) \Delta x_i(0) \rangle = \langle \Delta x_i(-t) \Delta x_i(0) \rangle$$

hence, autocorrelation is an even function.

1.2 Symmetry Properties

Let the generalized coordinates with positions q_i and associated momenta p_i be denoted by

$$\Gamma = (q_1, q_2, \dots, q_f, p_1, p_2, \dots, p_f)$$

Then, we make the time reversal transformation as:

$$q_i \rightarrow q_i, \quad t \rightarrow -t, \quad p_i \rightarrow -p_i$$

$$\Gamma \rightarrow -\Gamma = (q_1, \dots, q_f, -p_1, \dots, -p_f)$$

Under this transformation, the Hamiltonian satisfies:

$$H(\Gamma) = H(-\Gamma)$$

Now, observables typically transform as:

$$\hat{x}_i(-\Gamma) = \epsilon_i \hat{x}_i(\Gamma), \quad \epsilon_i \in \{+1, -1\}$$

- Position / energy: $\epsilon_i = +1$
- Velocity / angular momentum: $\epsilon_i = -1$

Hence, under time reversal:

$$\langle \Delta x_i(t) \Delta x_j(0) \rangle = \epsilon_i \epsilon_j \langle \Delta x_i(-t) \Delta x_j(0) \rangle$$

If $\epsilon_i \epsilon_j = 1$, then:

$$\langle \Delta x_i(t) \Delta x_j(0) \rangle = \langle \Delta x_i(-t) \Delta x_j(0) \rangle$$

Thus, the cross-correlation function too becomes even. This with stationarity at equilibrium also gives the $i \leftrightarrow j$ symmetry (adding t to both arguments):

$$\langle \Delta x_i(t) \Delta x_j(0) \rangle = \langle \Delta x_j(t) \Delta x_i(0) \rangle$$

1.3 Displacement Correlation

We distinguish between:

$$\Delta x_i(t) = x_i(t) - \bar{x}_i \quad (\text{deviation})$$

$$x_i(t) - x_i(0) \quad (\text{displacement})$$

Using:

$$x_i(t) - x_i(0) = \Delta x_i(t) - \Delta x_i(0)$$

we write:

$$\langle (x_i(t) - x_i(0))(x_j(t) - x_j(0)) \rangle = \langle (\Delta x_i(t) - \Delta x_i(0))(\Delta x_j(t) - \Delta x_j(0)) \rangle$$

Expanding:

$$= \langle \Delta x_i(t) \Delta x_j(t) \rangle + \langle \Delta x_i(0) \Delta x_j(0) \rangle - \langle \Delta x_i(t) \Delta x_j(0) \rangle - \langle \Delta x_j(t) \Delta x_i(0) \rangle$$

Assuming that $\langle \Delta x_i(0)^2 \rangle$ exists. Using time translational invariance and $i \leftrightarrow j$ symmetry:

$$\langle \Delta x_i(t) \Delta x_j(0) \rangle = \langle \Delta x_j(t) \Delta x_i(0) \rangle$$

Thus:

$$\langle (x_i(t) - x_i(0))(x_j(t) - x_j(0)) \rangle = 2 [\langle \Delta x_i(0) \Delta x_j(0) \rangle - \langle \Delta x_i(t) \Delta x_j(0) \rangle]$$

1.4 Velocity Correlation

The velocity associated with coordinate x_i is:

$$\dot{x}_i(t) = \frac{dx_i}{dt}$$

Note: $\frac{d}{dt}$ and $\langle \cdot \rangle$ commute.

$$\langle \dot{x}_i(t) \dot{x}_j(t') \rangle = \frac{\partial^2}{\partial t \partial t'} \langle \Delta x_i(t - t') \Delta x_j(0) \rangle$$

Substituting $t' = 0$:

$$\langle \dot{x}_i(t) \dot{x}_j(0) \rangle = -\frac{d^2}{dt^2} \langle \Delta x_i(t) \Delta x_j(0) \rangle$$

Using the previously derived relation between deviation and displacement correlations, we may equivalently express results in terms of displacement correlation ' $S(t)$ '.

$$S(t) = \langle (x_i(t) - x_i(0))(x_j(t) - x_j(0)) \rangle$$

Using the earlier relation between deviation and displacement correlations:

$$C_v(t) = \langle \dot{x}_i(t) \dot{x}_j(0) \rangle = \frac{1}{2} \frac{d^2 S(t)}{dt^2}$$

Integrating once:

$$2 \int_0^{t_1} C_v(t) dt = \frac{dS(t_1)}{dt} - \frac{dS(0)}{dt}$$

Since $S(t)$ is even:

$$\frac{dS}{dt}(0) = 0$$

Thus:

$$2 \int_0^{t_1} C_v(t) dt = \frac{dS(t_1)}{dt}$$

Integrating again:

$$2 \int_0^t dt_1 \int_0^{t_1} C_v(t_2) dt_2 = S(t) - S(0)$$

But:

$$S(0) = \langle (x_i(0) - x_i(0))(x_j(0) - x_j(0)) \rangle = 0$$

Hence:

$$S(t) = 2 \int_0^t dt_1 \int_0^{t_1} C_v(t_2) dt_2$$

1.5 Brownian Motion of a Free Particle

For a spherical particle, the drag coefficient in a fluid of viscosity η is given by Stokes' Law:

$$\zeta = 6\pi\eta R$$

Solving for the dynamics of a free spherical particle in such a fluid:

$$m \frac{dv}{dt} = -\zeta v$$

$$v(t) = v_0 e^{-t/\tau_v}, \quad \tau_v = \frac{m}{\zeta}$$

Stokes' law treats the surrounding fluid as a continuum, an approximation that is valid when the suspended particle is much larger than the fluid molecules. The average effect of an enormous number of collisions with the fluid molecules then appears as a viscous drag proportional to the particle's velocity. However, for particles that are sufficiently small (while still large enough for Stokes' law to remain valid), such as micron-sized beads or viruses, thermal fluctuations of the surrounding fluid molecules impart appreciable accelerations. As the velocity of the particle cannot then simply decay to zero due to the heat bath, the inclusion of a fluctuating force in addition to the average drag becomes necessary to describe the dynamics. We account for these fluctuations by introducing a random ("stochastic") force arising from the continual collisions with the fluid molecules, giving the **Langevin equation** (for a free particle):

$$m \frac{dv}{dt} = -\zeta v + F_r(t)$$

1.6 Properties of the Random Force

For a free Brownian particle, the isotropy of the system forces:

$$\langle v \rangle = 0, \quad \langle F_r(t) \rangle = 0$$

Random Force Correlation:

We model the random force as a sum of impulsive kicks:

$$F_r(t) = \sum_k p_k \delta(t - t_k)$$

where p_k denotes the impulse in each kick. We use Dirac delta functions to model the kicks, as molecular collisions occur over very short time scales (of the order of picoseconds), much smaller than the time scales of observation. Consider:

$$F_r(t_1)F_r(t_2) = \sum_{k,j} p_k p_j \delta(t_1 - t_k) \delta(t_2 - t_j)$$

Taking ensemble average:

$$\langle F_r(t_1)F_r(t_2) \rangle = \sum_{k,j} \langle p_k p_j \rangle \delta(t_1 - t_k) \delta(t_2 - t_j)$$

Since $\langle p_k \rangle = 0$ and the kicks are independent:

$$\langle p_k p_j \rangle = \langle p_k \rangle \langle p_j \rangle = 0 \quad (k \neq j)$$

Thus, only $k = j$ terms survive:

$$\langle p_k p_j \rangle = A' \delta_{kj}$$

Hence:

$$\langle F_r(t_1)F_r(t_2) \rangle = A' \sum_k \delta(t_1 - t_k) \delta(t_2 - t_k)$$

Using:

$$\delta(t_1 - t_k) \delta(t_2 - t_k) = \delta(t_1 - t_2) \delta(t_1 - t_k)$$

we get:

$$\langle F_r(t_1)F_r(t_2) \rangle = A' \delta(t_1 - t_2) \sum_k \delta(t_1 - t_k)$$

Now,

$$\sum_k \delta(t - t_k)$$

represents the density of collision events (as integrating over a time interval gives the number of kicks). Hence, in the continuum limit of rapidly occurring kicks, we use:

$$\sum_k \delta(t - t_k) \rightarrow \lambda$$

where λ is the average collision rate. Thus, $A = A'\lambda$ gives:

$$\boxed{\langle F_r(t_1)F_r(t_2) \rangle = A\delta(t_1 - t_2)}$$

Note: This derivation is pedagogical and not exact. In the continuum limit of rapidly occurring, independent (and hence uncorrelated in time) kicks, the force behaves as white noise, for which the correlation is proportional to $\delta(t_1 - t_2)$.

1.7 ‘Solving’ the Langevin Equation

$$m \frac{dv}{dt} = -\frac{m}{\tau_v} v + F_r(t)$$

Using the integrating factor $\mu(t) = e^{t/\tau_v}$:

$$\begin{aligned} e^{t/\tau_v} \frac{dv}{dt} + \frac{e^{t/\tau_v}}{\tau_v} v &= \frac{F_r(t)}{m} e^{t/\tau_v} \\ \frac{d}{dt} (v e^{t/\tau_v}) &= \frac{F_r(t)}{m} e^{t/\tau_v} \end{aligned}$$

Integrating from $-\infty$ to t :

$$v(t) = \frac{1}{m} \int_{-\infty}^t dt_1 e^{-(t-t_1)/\tau_v} F_r(t_1)$$

Determining the Constant A

Hence, the variance $\langle v(t)^2 \rangle$ is given by:

$$\langle v(t)^2 \rangle = \frac{1}{m^2} \int_{-\infty}^t dt_1 \int_{-\infty}^t dt_2 e^{-(t-t_1)/\tau_v} e^{-(t-t_2)/\tau_v} \langle F_r(t_1)F_r(t_2) \rangle$$

Using $\langle F_r(t_1)F_r(t_2) \rangle = A\delta(t_1 - t_2)$ as discussed in previous section:

$$\begin{aligned} \langle v(t)^2 \rangle &= \frac{A}{m^2} \int_{-\infty}^t dt_1 e^{-2(t-t_1)/\tau_v} \\ &= \frac{A}{m^2} \left[\frac{\tau_v}{2} e^{-2(t-t_1)/\tau_v} \right]_{-\infty}^t = \frac{A\tau_v}{2m^2} \end{aligned}$$

By the Equipartition Theorem for a 1D case (only one degree of freedom):

$$\langle v(t)^2 \rangle = \frac{k_B T}{m}$$

Equating the two expressions:

$$\frac{k_B T}{m} = \frac{A\tau_v}{2m^2} \implies A = \frac{2mk_B T}{\tau_v} = 2\zeta k_B T$$

Thus, the correlation function for the random force becomes:

$$\boxed{\langle F_r(t_1)F_r(t_2) \rangle = 2\zeta k_B T \delta(t_1 - t_2)}$$

Here, $t_1 = t_2$ gives an infinite variance for $F_r(t)$, as expected in our idealized model ($\delta(0) \rightarrow \infty$). For an exact characterization, we treat $F_r(t)$ as a Gaussian random variable with zero mean and the derived variance. By the Central Limit Theorem (CLT), many independent random collisions by fluid particles lead to a Gaussian distribution of force. Hence, F_r follows **Gaussian White Noise**.

1.8 Velocity Autocorrelation Function

To calculate $\langle v(t)v(t') \rangle$:

$$\langle v(t)v(t') \rangle = \frac{2\zeta k_B T}{m^2} \int_{-\infty}^t dt_1 \int_{-\infty}^{t'} dt_2 e^{-(t-t_1)/\tau_v} e^{-(t'-t_2)/\tau_v} \delta(t_1 - t_2)$$

The delta function $\delta(t_1 - t_2)$ forces the contribution to occur only when $t_1 = t_2 = s$. Without loss of generality (WLOG), let $t > t'$. Then s must lie within the overlapping domain of integration, i.e., $s \in (-\infty, \min(t, t')) \implies s \in (-\infty, t')$.

$$\begin{aligned} \langle v(t)v(t') \rangle &= \frac{2\zeta k_B T}{m^2} \int_{-\infty}^{t'} e^{-(t+t'-2s)/\tau_v} ds \\ &= \frac{2\zeta k_B T}{m^2} e^{-(t+t')/\tau_v} \left[\frac{\tau_v}{2} e^{2s/\tau_v} \right]_{-\infty}^{t'} \\ &= \frac{2\zeta k_B T}{m^2} \frac{\tau_v}{2} e^{-(t+t')/\tau_v} e^{2t'/\tau_v} \\ &= \frac{k_B T}{m} e^{-(t-t')/\tau_v} \end{aligned}$$

To make this expression general for any ordering of t and t' , we replace $t - t'$ with its absolute value:

$$\boxed{\langle v(t)v(t') \rangle = \frac{k_B T}{m} \exp\left(-\frac{|t-t'|}{\tau_v}\right)}$$

1.9 Mean-Squared Displacement and Diffusion Coefficient

We know that the displacement correlation can be related to the velocity correlation via:

$$\langle (x(t) - x(0))^2 \rangle = 2 \int_0^t dt_1 \int_0^{t_1} dt_2 \langle v(t_1)v(t_2) \rangle$$

Let $\tau_v \ll t$ (large system timescale, meaning the particle loses memory of its initial velocity rapidly). For characteristic observations where $t_1 \gg \tau_v$:

$$\int_0^{t_1} dt_2 \langle v(t_1)v(t_2) \rangle \approx \int_0^\infty dt_2 \langle v(t_2)v(0) \rangle$$

This approximation holds because the exponential decay prevents values where $t_2 \gg \tau_v$ from contributing significantly. We define the **self-diffusion constant** D as:

$$D = \int_0^\infty dt \langle v(t)v(0) \rangle$$

Therefore, the mean-squared displacement is (replacing t by $|t|$ to account for even symmetry):

$$\langle (x(t) - x(0))^2 \rangle = 2D|t|$$

By executing the explicit integration using our velocity correlation expression:

$$D = \frac{k_B T}{m} \tau_v \Rightarrow \boxed{D = \frac{k_B T}{\zeta}} ; \text{ This boxed expression is known as the } \mathbf{Einstein Relation}.$$

Here, τ_v acts as the velocity correlation time. In soft matter limits, we often treat $\tau_v \rightarrow 0$, which simplifies the velocity correlation structure to:

$$\langle v(t)v(t') \rangle = 2D\delta(t-t')$$

since $\frac{1}{2\tau_v} \exp\left(-\frac{|t-t'|}{\tau_v}\right) \rightarrow \delta(t-t')$. The factor of 2 appears because of the even symmetry of the autocorrelation function:

$$D = \frac{1}{2} \int_{-\infty}^\infty \langle v(t)v(0) \rangle dt$$

which is consistent with $\int_{-\infty}^\infty \delta(t) dt = 1$.

1.10 Brownian Motion in a Potential Field

When the particle is subjected to an external potential $U(x)$, the Langevin equation becomes:

$$m\ddot{x} = -\zeta\dot{x} - \frac{\partial U}{\partial x} + F_r(t)$$

Dividing through by ζ and using $\tau_v = \frac{m}{\zeta}$:

$$\tau_v\ddot{x} = -\dot{x} - \frac{1}{\zeta}\frac{\partial U}{\partial x} + v_r(t)$$

In the soft matter measurement timescales, the velocity relaxation time is negligible ($\tau_v \rightarrow 0$ or equivalently $m \rightarrow 0$), the inertia (or acceleration) term disappears. Therefore, over these coarse-grained observation timescales, any acceleration is rapidly dampened out through viscous fluid collisions (the relaxation of velocity), meaning there is no effective acceleration. Thus, the equation of motion reduces directly to:

$$\frac{dx}{dt} = -\frac{1}{\zeta}\frac{\partial U}{\partial x} + v_r(t)$$

where $v_r(t) = F_r(t)/\zeta$ denotes the random velocity component. Its properties are directly inherited from $F_r(t)$:

$$\langle v_r(t) \rangle = 0$$

$$\langle v_r(t)v_r(t') \rangle = \frac{1}{\zeta^2} \langle F_r(t)F_r(t') \rangle = \frac{2\zeta k_B T}{\zeta^2} \delta(t - t') = 2D\delta(t - t')$$

The random variable $v_r(t)$ remains Gaussian. For a sanity check, consider the fact that we expect the fluctuating component of velocity v_r to be a property of the fluid at a given temperature, and hence to be independent of the potential. Hence, choosing $U = 0$ (no potential, i.e., a free particle), we get $v_{\text{free}} = v_r$, which yields:

$$\langle v_r(t)v_r(t') \rangle = \langle v_{\text{free}}(t)v_{\text{free}}(t') \rangle = 2D\delta(t - t')$$

which was found earlier by explicitly solving for v_{free} in the short-relaxation ($\tau_v \rightarrow 0$) limit. Hence, showing consistency with above derivation.

The Harmonic Potential

Consider a simple harmonic potential $U(x) = \frac{1}{2}kx^2$. The dynamical equation becomes:

$$v = -\frac{k}{\zeta}x + v_r(t) = -\frac{x}{\tau} + v_r(t)$$

where $\tau = \zeta/k$. This differential equation has an identical form to the potential-free Langevin equation (up to parameter constants and v replaced by x). Hence, we can compute position correlation values $\langle x(t_1)x(t_2) \rangle$ using the integrating factor technique. Choosing the integrating factor IF = $e^{t/\tau}$:

$$\dot{x}e^{t/\tau} + \frac{x}{\tau}e^{t/\tau} = v_r(t)e^{t/\tau}$$

$$\frac{d}{dt} \left(x(t)e^{t/\tau} \right) = v_r(t)e^{t/\tau}$$

Integrating directly from the initial reference time 0 to a future time t :

$$\left[x(t')e^{t'/\tau} \right]_0^t = \int_0^t v_r(t_1)e^{t_1/\tau} dt_1$$

$$x(t)e^{t/\tau} - x_0 = \int_0^t v_r(t_1)e^{t_1/\tau} dt_1$$

where $x_0 := x(0)$. Thus we get $x(t)$ as:

$$x(t) = x_0 e^{-t/\tau} + \int_0^t v_r(t_1) e^{-(t-t_1)/\tau} dt_1$$

Now, to evaluate the position correlation function $\langle x(t)x(0) \rangle$ for $t > 0$, we multiply this expression by x_0 and take the ensemble average:

$$\langle x(t)x(0) \rangle = \langle x(t)x_0 \rangle = \left\langle \left(x_0 e^{-t/\tau} + \int_0^t v_r(t_1) e^{-(t-t_1)/\tau} dt_1 \right) x_0 \right\rangle$$

Distributing the expectation value yields two distinct terms:

$$\langle x(t)x(0) \rangle = \langle x_0^2 \rangle e^{-t/\tau} + \int_0^t \langle v_r(t_1)x_0 \rangle e^{-(t-t_1)/\tau} dt_1$$

To find the inner cross-correlation term $\langle v_r(t_1)x_0 \rangle$, we express the initial position x_0 by integrating from $-\infty$ to 0 instead of 0 to t :

$$x_0 = \int_{-\infty}^0 v_r(t_2) e^{t_2/\tau} dt_2$$

Substituting this definition into the cross-correlation bracket yields an inner integral over t_2 :

$$\langle v_r(t_1)x_0 \rangle = \int_{-\infty}^0 \langle v_r(t_1)v_r(t_2) \rangle e^{t_2/\tau} dt_2 = 2D \int_{-\infty}^0 \delta(t_1 - t_2) e^{t_2/\tau} dt_2$$

We evaluate the behavior of this inner integral over the domain of the outer integration variable $t_1 \in [0, t]$:

- For $t_1 > 0$: Because the inner integration bounds restrict t_2 to the past domain $(-\infty, 0]$, the $\delta(t_1 - t_2)$ spike lies entirely outside the integration region. Thus, $\langle v_r(t_1)x_0 \rangle = 0$ everywhere on the open interval $t_1 \in (0, t]$.
- For $t_1 = 0$: The delta function spikes exactly on the upper boundary of the integration domain ($t_2 = 0$). Evaluating a Dirac delta exactly on a boundary captures half of its total area, yielding an isolated finite value: $\langle v_r(0)x_0 \rangle = 2D \left(\frac{1}{2}e^0\right) = D$.

Because $\langle v_r(t_1)x_0 \rangle$ is perfectly zero everywhere except at the single isolated point $t_1 = 0$, the entire integral term vanishes completely, leaving only the exponential decay of the initial variance:

$$\int_0^t \langle v_r(t_1)x_0 \rangle e^{-(t-t_1)/\tau} dt_1 = 0 \Rightarrow \boxed{\langle x(t)x(0) \rangle = \langle x_0^2 \rangle e^{-t/\tau}}$$

Using the **Boltzmann distribution** ($P(x_0 = x) \propto e^{-U(x)/k_B T}$) for a particle in a harmonic potential $U(x) = \frac{1}{2}kx^2$, gives (via explicit integration):

$$\frac{1}{2}k\langle x_0^2 \rangle = \frac{1}{2}k_B T \implies \langle x_0^2 \rangle = \frac{k_B T}{k}$$

Substituting this variance back into our remaining term and employing the absolute value $|t|$ to account for time-reversal symmetry (evenness) yields the final, universal position correlation function:

$$\boxed{\langle x(t)x(0) \rangle = \frac{k_B T}{k} e^{-|t|/\tau}}$$

Hence, $\tau = \zeta/k$ represents position relaxation time. Then, the displacement correlation becomes

$$\langle (x(t) - x(0))^2 \rangle = 2 [\langle x(0)^2 \rangle - \langle x(t)x(0) \rangle] = \frac{2k_B T}{k} (1 - e^{-|t|/\tau}).$$

Now, in the regime $\tau_v \ll t \ll \tau$, we have:

$$e^{-|t|/\tau} \approx 1 - \frac{|t|}{\tau}.$$

Therefore,

$$\boxed{\langle (x(t) - x(0))^2 \rangle \approx \frac{2k_B T}{k} \frac{|t|}{\tau} = 2D|t|}$$

We get the velocity autocorrelation from $S(t) = \langle (x(t) - x(0))^2 \rangle$ by:

$$C_v(t) = \langle \dot{x}(t)\dot{x}(0) \rangle = \frac{1}{2} \frac{d^2 S(t)}{dt^2}.$$

Using

$$S(t) = \frac{2k_B T}{k} (1 - e^{-|t|/\tau}),$$

we first compute the first derivative. Since $\frac{d|t|}{dt} = \text{sgn}(t)$, we get:

$$\frac{d}{dt} e^{-|t|/\tau} = -\frac{1}{\tau} e^{-|t|/\tau} \text{sgn}(t)$$

Therefore,

$$\frac{dS}{dt} = \frac{2k_B T}{k\tau} e^{-|t|/\tau} \text{sgn}(t)$$

Differentiating again:

$$\frac{d^2 S}{dt^2} = \frac{2k_B T}{k\tau} \frac{d}{dt} \left(e^{-|t|/\tau} \text{sgn}(t) \right)$$

Using

$$\text{sgn}(t) = \theta(t) - \theta(-t),$$

we write

$$e^{-|t|/\tau} \text{sgn}(t) = e^{-t/\tau} \theta(t) - e^{t/\tau} \theta(-t).$$

Hence,

$$\frac{d^2 S}{dt^2} = \frac{2k_B T}{k\tau} \frac{d}{dt} \left[e^{-t/\tau} \theta(t) - e^{t/\tau} \theta(-t) \right]$$

Differentiating term by term:

$$\frac{d}{dt} \left(e^{-t/\tau} \theta(t) \right) = -\frac{1}{\tau} e^{-t/\tau} \theta(t) + e^{-t/\tau} \delta(t),$$

and

$$\frac{d}{dt} \left(e^{t/\tau} \theta(-t) \right) = \frac{1}{\tau} e^{t/\tau} \theta(-t) - e^{t/\tau} \delta(t)$$

Thus,

$$\frac{d^2 S}{dt^2} = \frac{2k_B T}{k\tau} \left[e^{-t/\tau} \delta(t) + e^{t/\tau} \delta(t) - \frac{1}{\tau} e^{-|t|/\tau} \right].$$

$$e^{-0/\tau} + e^{0/\tau} = 2 \quad \Rightarrow \quad \frac{d^2 S}{dt^2} = \frac{2k_B T}{k\tau} \left[2\delta(t) - \frac{1}{\tau} e^{-|t|/\tau} \right].$$

Therefore,

$$C_v(t) = \frac{1}{2} \frac{d^2 S}{dt^2} = \frac{k_B T}{k\tau} \left[2\delta(t) - \frac{1}{\tau} e^{-|t|/\tau} \right].$$

Using

$$D = \frac{k_B T}{\zeta}, \quad \tau = \frac{\zeta}{k},$$

we note that

$$\frac{k_B T}{k\tau} = D,$$

Hence,

$$C_v(t) = D \left[2\delta(t) - \frac{e^{-|t|/\tau}}{\tau} \right]$$

As $t \ll \tau$, the first term dominates the second and we approximate:

$$\boxed{C_v(t) = 2D\delta(t)}$$

In the regime $\tau_v \ll t \ll \tau$, the particle has already lost memory of its initial velocity due to viscous damping, but has not yet moved sufficiently far for the confining potential to significantly influence its motion. Over such short positional displacements, the variation of the potential experienced by the particle is negligible. Hence, the dynamics locally resembles that of a free Brownian particle. Therefore, the correlation functions reduce to the same forms obtained earlier for the potential-free case. Note that the inequality $\tau_v \ll \tau$ assumed here holds for most ordinary Brownian systems, though exceptions can arise in weakly dissipative, strongly confined, or highly inertial systems (as $\tau_v > \tau \iff mk > \zeta^2$).

2 Itô calculus and the Fokker-Planck equation

2.1 The Wiener Process

The Langevin equation can be seen as a manifestation of the general stochastic differential equation (SDE):

$$\frac{dX}{dt} = \mu(X, t) + \sigma(X, t)\xi(t)$$

Here, $\xi(t)$ is delta correlated Gaussian white noise, that is

$$\langle \xi(t)\xi(t') \rangle = \delta(t - t'), \quad \langle \xi(t) \rangle = 0$$

Due to the singular (infinite variance) nature of $\xi(t)$, we seek a better representation for this SDE. For this, we discretize the time axis to get a non-singular variance for the noise which becomes exact in the limit $\Delta t \rightarrow 0$, using a step-size Δt , so the time points are given by $t_n = n\Delta t$ ($n \in \mathbb{Z}$). Then, integrals of a function f are given by a Riemann sum (which again becomes exact in the limit $\Delta t \rightarrow 0$).

$$\int_{t_i}^{t_j} f(t)dt \approx \sum_{k=i}^j f(t_k)\Delta t$$

Thus, to satisfy the properties of $\delta(t)$, we define our discrete delta function by

$$\delta(t_i) = \delta_{i0}/\Delta t$$

where δ_{ij} is the Kronecker delta so it becomes a rectangular peak at the origin (so it becomes $\delta(t)$ in the limit $\Delta t \rightarrow 0$). This gives the noise correlation to be:

$$\langle \xi(t_i)\xi(t_j) \rangle = \delta(t_i - t_j) = \delta(t_{i-j}) = \frac{\delta_{(i-j)0}}{\Delta t} = \frac{\delta_{ij}}{\Delta t}$$

This gives the variance of the noise to be $\langle \xi(t_i)^2 \rangle = 1/\Delta t$. Therefore, to get a more well behaved quantity in the $\Delta t \rightarrow 0$ limit, we define:

$$W(t_i) = \int_0^t \xi(t')dt' \approx \sum_{k=0}^i \xi(t_k)\Delta t$$

Note that in the limit $\Delta t \rightarrow 0$, an arbitrarily large choice of i allows t_i to approach any real time t . As $\xi(t) \sim \mathcal{N}(0, 1/\Delta t)$, $\xi(t)\Delta t \sim \mathcal{N}(0, \Delta t)$ we get:

$$\Delta W(t_i) := W(t_i) - W(t_{i-1}) = \xi(t_i)\Delta t \sim \mathcal{N}(0, \Delta t)$$

Thus, ΔW becomes on the order of $\sqrt{\Delta t}$ (its standard deviation) and hence infinitesimal in the limit $\Delta t \rightarrow 0$. In fact, $W(t)$ is continuous at all points (almost surely, i.e. with probability 1) as expected from an infinitesimal differential (though the rigorous proof lies beyond our scope). Further an increment in W for two times t_i to t_j is given by:

$$W(t_j) - W(t_i) = \sum_{k=i+1}^j \xi(t_k)\Delta t = \sum_{k=i+1}^j \Delta W(t_k)$$

As we know $\Delta W(t_k) \sim \mathcal{N}(0, \Delta t)$ and that summing independent normal distributions adds up their variances and means to give another normal variate, we get $W(t_j) - W(t_i) \sim \mathcal{N}(0, |j - i|\Delta t) = \mathcal{N}(0, |t_j - t_i|)$. Therefore, in the continuum limit with $\Delta t \rightarrow 0$, we have:

$$W(t) - W(s) \sim \mathcal{N}(0, |t - s|)$$

for any two times t and s . We also see that $W(t)$ inherits its mean directly from $\xi(t)$:

$$\langle W(t) \rangle = \int_0^t \langle \xi(t') \rangle dt' = \int_0^t 0 dt' = 0$$

Further, by the definition of an integral, we have:

$$W(0) = \int_0^0 \xi(t)dt' = 0$$

We may also find the correlation (called covariance in more mathematical terms) between two instances of W as:

$$\langle W(t)W(s) \rangle = \int_0^t \int_0^s \langle \xi(t')\xi(s') \rangle dt' ds' = \int_0^t \int_0^s \delta(t' - s') dt' ds' = \min(t, s)$$

where t and s are positive, as we typically only care about time elapsed from an initial zero, this is usually taken to be the domain for $W(t)$. Finally, as the noise-kicks are independent at different times, increments $[W(b) - W(a)]$ and $[W(d) - W(c)]$ for some $0 < a < b < c < d$ are independent (i.e. increments in $W(t)$ for non-overlapping pair of intervals are independent).

The discussed properties of $W(t)$ fully characterize it as a stochastic process. $W(t)$ is called the **Wiener Process**. Given its (relatively) well behaved properties, we intend to rewrite our SDE in terms of it. To do this, we take the continuum

limit ($\Delta t \rightarrow 0$), where we may replace $\Delta W(t_i) = \xi(t_i)\Delta t$ by $dW = \xi(t)dt$, therefore, we rewrite our SDE in the form of differentials, called an **Itô SDE**:

$$\boxed{dX = \mu dt + \sigma dW}$$

It is worth noting that in rigorous stochastic calculus, the Wiener process is defined with the properties we motivated above without any concern for δ -correlated noise, which is a singular object representing a distributional derivative of $W(t)$ (which doesn't exist in the usual sense as $\Delta W/\Delta t \sim 1/\sqrt{\Delta t}$). We shall now replace Δt by dt to denote the continuum limit, then we have $dW \sim \mathcal{N}(0, dt)$, which gives

$$\langle dX|X(t) = x \rangle = \mu(x, t)dt + \sigma(x, t)\langle dW \rangle = \mu(x, t)dt \implies \frac{\langle dX|X(t) = x \rangle}{dt} = \mu(x, t)$$

Thus, $\mu(x, t)$ represents the average rate of change of X at given 'position' x and 'time' t and is therefore called the **drift** term. Further, as the variance of $W(t)$ increases with time, it causes X to 'diffuse', so we call $\sigma(x, t)$ the **diffusion** strength.

2.2 Itô's Lemma

Often, we may be interested in changes in some function $f(X, t)$ of the stochastic variable, like energy or as we'll see in the next section, probability distributions. Hence, we seek to write an Itô SDE for df , as we are only concerned with first order behavior in time (to know the rate of change of f), we use mean value theorem to get the following:

$$f(X + \Delta X, t + \Delta t) \approx f(X + \Delta X, t) + \Delta t \frac{\partial f}{\partial t}(X + \Delta X, t') \quad (t' \in [\min(t, t + \Delta t), \max(t, t + \Delta t)])$$

Now, as $\Delta W \sim \sqrt{\Delta t}$ and ΔX has a term involving it, we cannot ignore the second order terms as $(dW)^2 \sim dt$. In fact, we note that

$$\text{Var}(\Delta W) = \langle (\Delta W)^2 \rangle = \Delta t$$

and $\text{Var}((\Delta W)^2) = \langle (\Delta W)^4 \rangle - \langle (\Delta W)^2 \rangle^2$. For normal random variable $Y \sim \mathcal{N}(\mu, \sigma^2)$, the fourth order raw moment is given by $E(Y^4) = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$, applying this to ΔW , we get:

$$\text{Var}((\Delta W)^2) = 3\langle (\Delta W)^2 \rangle^2 - \langle (\Delta W)^2 \rangle^2 = 2(\Delta t)^2$$

Therefore, in the limit $\Delta t \rightarrow 0$, $\text{Var}((\Delta W)^2) \rightarrow 0$ and $\langle (\Delta W)^2 \rangle = \Delta t$ so $(\Delta W)^2$ converges to Δt in probability (i.e. the probability of $(\Delta W)^2$ lying outside an arbitrarily small neighborhood of Δt approaches 0, this can be proved easily via Chebyshev's inequality). Therefore, we write:

$$dW^2 = dt$$

whereas terms like $dWdt$ and dt^2 can be ignored as they are of orders higher than dt and so vanish when summed in an integral. The above motivated $(dW)^2 = dt$ is in fact true in more stronger sense, i.e. for an arbitrary integrable function $g(t)$, the sums $\sum_{i=0}^N g(t_i)(\Delta W(t_i))^2$ and $\sum_{i=0}^N g(t_i)\Delta t$ converge to the same number almost surely (the first stochastic sum collapsing to a single point with probability 1), the proof of which we omit here to maintain focus. Thus, we use a Taylor expansion of second order in ΔX in our previous expression:

$$f(X + \Delta X, t + \Delta t) \approx f(X, t) + \Delta X \frac{\partial f}{\partial x}(X, t) + \frac{(\Delta X)^2}{2} \frac{\partial^2 f}{\partial x^2}(X, t) + \frac{\partial f}{\partial t}(X + \Delta X, t)\Delta t \quad (X' \in [\min(X, X + \Delta X), \max(X, X + \Delta X)])$$

Now, taking the limit $\Delta X \rightarrow 0$ and $\Delta t \rightarrow 0$, we get $X' \rightarrow X, t' \rightarrow t$ and $f(X + \Delta X, t + \Delta t) - f(X, t)$ becomes df , so assuming continuity of derivatives, we get:

$$df = dX \frac{\partial f}{\partial x}(X, t) + \frac{(dX)^2}{2} \frac{\partial^2 f}{\partial x^2}(X, t) + dt \frac{\partial f}{\partial t}(X, t)$$

Substituting dX from original SDE, we get:

$$df = (\mu dt + \sigma dW) \frac{\partial f}{\partial x} + \frac{(\mu dt + \sigma dW)^2}{2} \frac{\partial^2 f}{\partial x^2} + dt \frac{\partial f}{\partial t}$$

Using the rules $dt \cdot dW = 0$, $(dW)^2 = dt$ and $(dt)^2 = 0$ discussed earlier, we get **Itô's Lemma**:

$$\boxed{df = \left(\mu \frac{\partial f}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} + \frac{\partial f}{\partial t} \right) dt + \sigma \frac{\partial f}{\partial x} dW}$$

Here, we the only restriction on f used is that $f \in C^{2,1}$ (continuously differentiable twice in 'position' and continuously differentiable once in 'time').

2.3 The Fokker-Planck Equation

In statistical mechanics, we are often interested in how the probability distribution of $X(t)$ for an ensemble of particles evolves rather than the exact trajectory of a Brownian particle governed by the Langevin equation. To find this evolution law, we note that given a function $f(X)$ of the stochastic variable, its expectation value is given by:

$$\langle f(X) \rangle = \int_{-\infty}^{\infty} f(x) P(x, t) dx$$

where $P(x, t)$ is the probability density at $X(t) = x$. Therefore, we aim to use Itô's Lemma to find an expression for $\partial P / \partial t$ by having f act as an arbitrary test function (smooth function with compact support). Choosing f with no explicit time dependence, we have $\partial f / \partial t = 0$, so Itô's lemma gives:

$$df = \left(\mu \frac{\partial f}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW$$

Taking the expectation value of both sides, as $dW = \xi(t)dt$ is independent from $\sigma(X, t) \frac{\partial f}{\partial x}(X)$ and $\langle dW \rangle = 0$, we get $\langle \sigma \frac{\partial f}{\partial x} dW \rangle = \langle \sigma \frac{\partial f}{\partial x} \rangle \langle dW \rangle = 0$, which gives:

$$\langle df \rangle = \left\langle \mu \frac{\partial f}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} \right\rangle dt$$

As averaging is a linear operation (and a differential is the limit of a difference), $\langle df \rangle = d\langle f \rangle$ which gives:

$$\frac{d\langle f(X) \rangle}{dt} = \left\langle \mu \frac{\partial f}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} \right\rangle$$

Recalling the expression for expectation value, we get:

$$\int_{-\infty}^{\infty} f(x) \frac{\partial P}{\partial t} dx = \int_{-\infty}^{\infty} \left(\mu \frac{\partial f}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} \right) P(x, t) dx$$

Integrating the first term on the right-hand side by parts, we get:

$$\int_{-\infty}^{\infty} \mu \frac{\partial f}{\partial x} P(x, t) dx = [\mu \cdot P(x, t) f(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x) \frac{\partial}{\partial x} (\mu \cdot P(x, t)) dx = - \int_{-\infty}^{\infty} f(x) \frac{\partial}{\partial x} (\mu \cdot P(x, t)) dx$$

where the boundary term vanishes as f has compact support (vanishes outside a compact domain). By the same logic, $f'(x)$ also vanishes at infinity, which allows us to simplify the second term by integration by parts applied twice:

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} P(x, t) dx &= \left[f'(x) \frac{\sigma^2}{2} \cdot P(x, t) \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{f'(x)}{2} \frac{\partial(\sigma^2 P(x, t))}{\partial x} dx = - \int_{-\infty}^{\infty} \frac{f'(x)}{2} \frac{\partial(\sigma^2 P(x, t))}{\partial x} dx \\ \Rightarrow \int_{-\infty}^{\infty} \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} P(x, t) dx &= - \left[\frac{f(x)}{2} \frac{\partial(\sigma^2 P(x, t))}{\partial x} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{f(x)}{2} \frac{\partial^2(\sigma^2 P(x, t))}{\partial x^2} dx = \int_{-\infty}^{\infty} \frac{f(x)}{2} \frac{\partial^2(\sigma^2 P(x, t))}{\partial x^2} dx \end{aligned}$$

Substituting the integrals in the earlier expression, we get:

$$\int_{-\infty}^{\infty} f(x) \frac{\partial P}{\partial t} dx = \int_{-\infty}^{\infty} f(x) \left[\frac{1}{2} \frac{\partial^2(\sigma^2 P(x, t))}{\partial x^2} - \frac{\partial}{\partial x} (\mu \cdot P(x, t)) \right] dx$$

As the equality holds for all arbitrary test functions $f(x)$, we obtain:

$$\boxed{\frac{\partial P}{\partial t} = \frac{1}{2} \frac{\partial^2(\sigma^2 P(x, t))}{\partial x^2} - \frac{\partial}{\partial x} (\mu \cdot P(x, t))}$$

This boxed equation is known as the **Fokker-Planck equation**. We may rewrite it in the form of a continuity equation as:

$$\frac{\partial P}{\partial t} = - \frac{\partial J}{\partial x}$$

where

$$J(x, t) = \left[\mu \cdot P(x, t) - \frac{1}{2} \frac{\partial(\sigma^2 P(x, t))}{\partial x} \right]$$

denotes the ‘probability current’ at a point x at time t . The first term $\mu \cdot P(x, t)$ represents the flow of probability density due to the deterministic drift term in the Itô SDE. The second term can be further broken as:

$$-\frac{1}{2} \frac{\partial(\sigma^2 P)}{\partial x} = -\frac{\sigma^2}{2} \frac{\partial P}{\partial x} - \frac{P}{2} \frac{\partial \sigma^2}{\partial x}$$

The first term in this decomposition represents Fick’s law of diffusion, where the current flows from regions of high probability density to low ones, with the current being faster for higher diffusion strength σ . This diffusive current is also proportional to the diffusion strength squared σ^2 . The second term represents flow from regions of higher diffusion strength to lower. Although we shall only deal with conditions where σ is constant, the overdamped Langevin equation ($\tau_v \rightarrow 0$) gives an intuitive physical interpretation for each component of J :

$$\frac{dx}{dt} = -\frac{1}{\zeta} \frac{\partial U}{\partial x} + \sqrt{2D} \xi(t)$$

where the second term is obtained from the correlation $\langle v_r(t) v_r(t') \rangle = 2D \delta(t - t')$. Thus, $\mu = F_{\text{potential}}/\zeta$ indicates that the probability distribution is pushed along the conservative force which is resisted by drag. Also, as $\sigma^2 = 2D = 2k_B T/\zeta$ the term $\propto -\partial \sigma^2 / \partial x$ may be heuristically viewed as representing probability flow from regions of higher to lower D . However, this is only a heuristic and not a prediction for overdamped dynamics in thermal or viscosity gradients; when σ varies with X , additional corrections are required which we do not concern ourselves with in this text.

While the continuity equation allows us to add an arbitrary function of time t to J , the expression for J written earlier is special as it equals the probability flux found directly via consideration of transition probabilities between different values of x . This is already motivated by the fact that every term in J has a physical interpretation in this particular example. We postpone the proof of this uniqueness to the appendix.

3 Wielding the Fokker-Planck Equation

3.1 The Boltzmann Distributions

Equipped with the Fokker-Planck equation, we now aim to find the probability distributions of x and v at equilibrium. In general, the Langevin equation includes both stochastic variables x and v while we have only discussed the Fokker-Planck equation in 1D. Thus, we shall deal with special cases of the Langevin equation that can be written in terms of only position or velocity. In the last section, we saw that the overdamped Langevin equation is one such equation for position, with $\mu(x) = F_{\text{potential}}/\zeta$ and $\sigma = \sqrt{2D}$. At equilibrium, the probability distribution is stationary, i.e. $\partial P / \partial t = 0$, which gives:

$$\frac{\partial J}{\partial x} = 0 \Rightarrow J = -\frac{P_{\text{eq}}(x)}{\zeta} \frac{\partial U}{\partial x} - D \frac{\partial P_{\text{eq}}(x)}{\partial x} = C$$

where C is a constant with respect to position and by the time-independence of each term in J at equilibrium, also with respect to time. However, as discussed at the end of the previous section, J represents flow of probability to different states, which cannot have a preferred direction due to the time-reversal symmetry of the equilibrium dynamics, a condition called **detailed balance**. As we shall see in the appendix, this forces $J = 0$, which gives:

$$\frac{1}{P_{\text{eq}}(x)} \frac{\partial P_{\text{eq}}(x)}{\partial x} = -\frac{1}{D\zeta} \frac{\partial U}{\partial x} \Rightarrow \log(P_{\text{eq}}(x)) = -\frac{U(x)}{D\zeta} + C \Rightarrow \boxed{P_{\text{eq}}(x) \propto e^{-U(x)/k_B T}}$$

where we used $D\zeta = k_B T$ and C is determined by normalization of the probability density. For v , we use the Langevin equation for a free particle, where $\mu(v) = -v/\tau_v$ and $\sigma = \sqrt{2\zeta k_B T}/m$ as $\dot{v} = -v/\tau_v + F_r/m$. Setting $J = 0$ at equilibrium in velocity space then gives:

$$-\frac{\zeta v}{m} P_{\text{eq}}(v) - \frac{2\zeta k_B T}{2m^2} \frac{\partial P_{\text{eq}}(v)}{\partial v} = 0 \Rightarrow \frac{1}{P_{\text{eq}}(v)} \frac{\partial P_{\text{eq}}(v)}{\partial v} = -\frac{mv}{k_B T} \Rightarrow \log(P_{\text{eq}}(v)) = -\frac{mv^2}{2k_B T} + C \Rightarrow \boxed{P_{\text{eq}}(v) \propto e^{-\frac{mv^2}{2k_B T}}}$$

Thus, under the special conditions of an overdamped dynamics for x and a free particle for v , we have recovered their respective Boltzmann distributions from Langevin dynamics itself, making our discussions in earlier sections self consistent, such that the only assumption from thermodynamics we relied on was $\langle v^2 \rangle = k_B T/m$ to obtain the magnitude of the random forc. Indeed, these distributions are generally true and the joint distribution

$$P_{\text{eq}}(x, v) \propto \exp \left[-\left(\frac{mv^2}{2} + U(x) \right) / k_B T \right]$$

may be recovered from the 2D Fokker-Planck equation obtained from the general underdamped Langevin equation, which we have omitted here to maintain focus. However, due to the independence of marginal distributions $P_{\text{eq}}(x)$ and $P_{\text{eq}}(v)$ from m and $U(x)$ respectively, taking the limits $m \rightarrow 0$ and $U(x) \rightarrow 0$ does not affect the distributions which we assumed in our special cases.

3.2 The Kramers' Escape Problem

Consider a confining potential well with its minimum at $x = a$ with a cross-able barrier on the right with a maximum at $x = c$ followed by a slope down through which the particle may 'escape' the well. We seek to study the rate of this escape or equivalently the average time a particle takes to escape (reciprocals of each other). We shall study this in the regime where the barrier height $U_0 = [U(c) - U(a)]$ is much larger than thermal energy $k_B T$ in the overdamped ($\tau_v \rightarrow 0$) case. Although we study the escape over a potential barrier across space, the mathematical approach stands for processes like changing torsion angle in protein folding or some reaction coordinate in certain chemical reactions in solutions. First, we notice that the expression for probability current for the overdamped Langevin equation maybe simplified as follows:

$$J = \frac{-1}{\zeta} \left[P \frac{\partial U}{\partial x} + k_B T \frac{\partial P}{\partial x} \right] \implies J \exp\left(\frac{U}{k_B T}\right) = -D \frac{\partial}{\partial x} \left(P \cdot \exp\left(\frac{U}{k_B T}\right) \right)$$

Now, we note that in the limit $k_B T \ll U_0$, the expected time before the particle escapes, τ_{esc} should be much larger than the timescale on which the distribution at the well bottom locally relaxes to a local equilibrium proportional to a Boltzmann distribution, which is of the order of $\tau_{\text{relax}} = \zeta/U''(a)$ (by a second order approximation as a harmonic potential). Thus, on the timescales $\tau_{\text{relax}} \ll t \ll \tau_{\text{escape}}$, relaxation is effectively instantaneous and the distribution is quasi-static, i.e. $\partial P/\partial t \approx 0$ which by the continuity equation implies $\partial J/\partial x \approx 0$ which means J is approximately constant over small segments. We may then integrate the earlier derived expression from b to d treating the current across the barrier J_{barrier} as constant, such that $a < b < c < d$ (b and d are points on both sides of the barrier peak), which gives:

$$J_{\text{barrier}} \int_b^d \exp\left(\frac{U(x)}{k_B T}\right) dx = -D \left[P \exp\left(\frac{U}{k_B T}\right) \right]_b^d$$

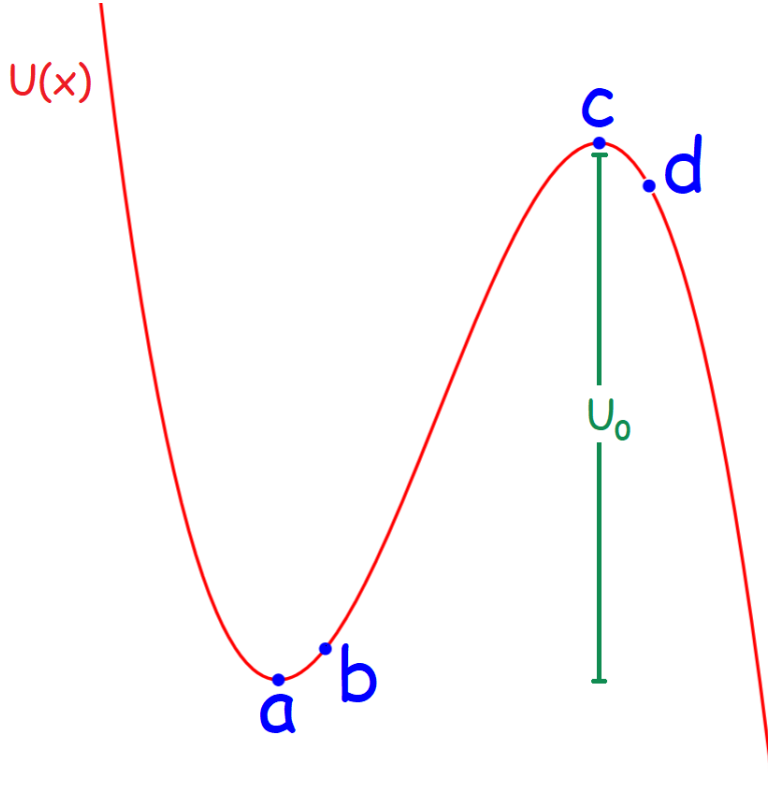


Figure: A schematic plot of the potential energy profile in the Kramers' escape problem, with barrier height U_0 , barrier peak c and well minimum a .

Now, we impose the conditions $k_B T \ll [U(b) - U(a)], [U(c) - U(d)] \ll U_0$ where the upper bound shall be used to ensure the validity of a quadratic approximation of the potential near a and c . The lower bound ensures b is further right in the well enough that nearly all the probability mass of the local Boltzmann quasi-equilibrium $P = A \exp(-U/k_B T)$ lies before it. We eliminate A using $P(a) \exp(U(a)/k_B T) = A$ so the total un-escaped probability can be given by:

$$P_{\text{well}} = \int_{-\infty}^b P(a) \exp\left(\frac{U(a) - U(x)}{k_B T}\right) dx$$

Using the quadratic approximation $U(a) - U(x) \approx -U''(a)(x-a)^2/2$ to evaluate this integral, we may also replace the upper limit by ∞ as b lies well beyond the peak of the integrand to get a standard Gaussian integral:

$$P_{\text{well}} \approx P(a) \int_{-\infty}^{\infty} \exp\left(\frac{-U''(a)(x-a)^2}{2k_B T}\right) dx = P(a) \sqrt{\frac{2\pi k_B T}{U''(a)}}$$

If the well continued without a barrier, the Boltzmann distribution would give $P(b) \exp(U(b)/k_B T) = P(d) \exp(U(d)/k_B T)$. However, the lower bound also ensures d lies well beyond the barrier so that the escaped density quickly drifts along the steep potential avoiding any accumulation at d making the term evaluated at d negligible compared to the one at b . We may also write $P(b) \exp(U(b)/k_B T) = P(a) \exp(U(a)/k_B T)$ by the local equilibrium allowing us to simplify the expression for J_{barrier} as:

$$J_{\text{barrier}} = D \cdot P(a) \exp\left(\frac{U(a)}{k_B T}\right) \bigg/ \int_b^d \exp\left(\frac{U(x)}{k_B T}\right) dx$$

As the integral is dominated by values near the barrier peak at c , we again use the quadratic approximation $U(x) \approx U(c) - |U''(c)|(x-c)^2/2 = U_0 + U(a) - |U''(c)|(x-c)^2/2$ and extend limits to $\pm\infty$ and evaluate a Gaussian integral:

$$\int_b^d \exp\left(\frac{U(x)}{k_B T}\right) \approx \exp\left(\frac{U(a) + U_0}{k_B T}\right) \int_{-\infty}^{\infty} \exp\left(\frac{-|U''(c)|(x-c)^2}{2k_B T}\right) = \exp\left(\frac{U(a) + U_0}{k_B T}\right) \sqrt{\frac{2\pi k_B T}{|U''(c)|}}$$

This gives the expression for probability current across the barrier to be:

$$J_{\text{barrier}} = D \cdot P(a) \exp\left(\frac{-U_0}{k_B T}\right) \sqrt{\frac{|U''(c)|}{2\pi k_B T}}$$

We can eliminate the variable $P(a)$ by combining the expressions for J_{barrier} and P_{well} to get:

$$J_{\text{barrier}} = D \cdot P_{\text{well}} \frac{\sqrt{U''(a)|U''(c)|}}{2\pi k_B T} \exp\left(\frac{-U_0}{k_B T}\right) = \exp\left(\frac{-U_0}{k_B T}\right) \frac{\sqrt{U''(a)|U''(c)|}}{2\pi\zeta} P_{\text{well}}$$

Thus, the current across the barrier is proportional to the density confined in the well, as expected on physical grounds. We can now expect the (Fokker-Planck) continuity equation to use this to give a differential equation for P_{well} as a function of time. Integrating the continuity equation from $-\infty$ to b , taking $J(-\infty) = 0$ (justified by the local Boltzmann distribution decaying along rising potential to the left) and $J(b) = J_{\text{barrier}}$ by approximate constancy of J across space as discussed before, we obtain:

$$\frac{\partial P_{\text{well}}}{\partial t} = -J(b) + J(-\infty) = -J_{\text{barrier}} = -\frac{\sqrt{U''(a)|U''(c)|}}{2\pi\zeta} \exp\left(\frac{-U_0}{k_B T}\right) P_{\text{well}}$$

At $t = 0$, as the particle is initially confined in the well, the probability for the particle to lie in the well region $(-\infty, b]$ is unity, i.e. $P_{\text{well}}(0) = 1$. Combined with the differential equation above, we obtain $P_{\text{well}}(t) = e^{-rt}$ where the decay rate r is given by:

$$r = \frac{\sqrt{U''(a)|U''(c)|}}{2\pi\zeta} \exp\left(\frac{-U_0}{k_B T}\right)$$

To better interpret r physically, we note that $P_{\text{well}}(t)$ equals the probability that the particle has not escaped the well till time t or equivalently $\text{Prob}(t < t_{\text{esc}})$ where t_{esc} is the time the particle took to escape the well, a random variable. Then, the cumulative probability distribution function for t_{esc} is $\text{Prob}(t_{\text{esc}} < t) = 1 - \text{Prob}(t < t_{\text{esc}}) = 1 - e^{-rt}$. This gives the probability distribution function for the escape time to be:

$$f(t) = \frac{d}{dt} (1 - e^{-rt}) = re^{-rt} \quad (t > 0)$$

Then, the average time a particle spends in the well before escaping becomes:

$$\tau_{\text{escape}} = \int_0^{\infty} t \cdot (re^{-rt}) dt = \frac{1}{r} \int_0^{\infty} ye^{-y} dy = \frac{1}{r}$$

Thus, the average escape time becomes:

$$\tau_{\text{escape}} = \frac{1}{r} = \frac{2\pi\zeta}{\sqrt{U''(a)|U''(c)|}} \exp\left(\frac{U_0}{k_B T}\right)$$

Thus, for an ensemble of particles starting in the well, r represents the rate of escape (number of particles escaping per unit time). We see that the rate of escape increases with increased curvature of the barrier peak and the well minimum and is inversely proportional to the viscosity or drag in the medium. Additionally, our application of the Fokker-Planck equation to this problem has also reproduced the famous exponential factor $\exp(-E_a/k_B T)$ in the Arrhenius equation for the rate of a chemical reaction where E_a represents the activation energy for a transition state.

4 Simulating Langevin Dynamics

4.1 The Underdamped Langevin Equation

The Langevin equation is said to be ‘underdamped’ when we do not ignore the mass/acceleration term (which is non negotiable when working on timescales on the order of τ_v). To implement this, we initialize the position and velocity of the particle by x_0 and v_0 at $t = 0$. To simulate the equation numerically, we need to discretize the Langevin equation. Thus, we update position and velocity of the particle at equal intervals of time (our timestep) ‘ dt ’ (fixed at start of simulation). For position, we use a second order Taylor approximation to get the following recurrence relation:

$$x_{n+1} = x_n + v_n dt + \frac{F_n (dt)^2}{m \cdot 2}$$

Here, F_n denotes the force on the particle at $t = t_n = n dt$ (n^{th} timestep). We write $F_n = F_{\text{drag}}(n) + F_{\text{harm}}(n) + F_r(n)$, where $F_{\text{drag}}(n) = -\zeta v_n$ denotes the drag force, $F_{\text{harm}}(n) = -kx_n$ denotes the force due to the harmonic potential and $F_r(n)$ is the random stochastic force.

To implement $F_r(n)$ as Gaussian noise, we use a normal distribution sample generator like `numpy.random.normal()` in python. The earlier derived correlation function gives the variance of this normal distribution to be $\langle F_r(n)^2 \rangle = 2\zeta k_B T \delta(0)$ which would be undefined if $\delta(t)$ is taken to be a singular spike as usual. Instead, in our discrete model, as the timescales we integrate over have the lower bound dt , we can replace it by a rectangle of width ‘ dt ’ and height ‘ $1/dt$ ’ centered at the origin (area of rectangle remains unity as required for the $\delta(t)$). This gives the variance of the normal distribution as $\delta(0) = 1/dt \implies \langle F_r(n)^2 \rangle = 2\zeta k_B T / dt$. Therefore, $F_r(n)$ is taken as a sample from a normal distribution of mean zero and standard deviation $\sqrt{2\zeta k_B T / dt}$ at every iteration.

As the recurrence relation for x_n (the position at $t = t_n$) involves v_n (the velocity at $t = t_n$), we need the corresponding relation for velocity. To obtain a second order approximation of v_n , it suffices to approximate acceleration $a_n = F_n/m$ to the first order in, i.e. $a_{n+1} = a_n + \dot{a}_n dt \Rightarrow \dot{a}_n = (a_{n+1} - a_n)/dt$, which gives:

$$v_{n+1} = v_n + a_n dt + \dot{a}_n \frac{(dt)^2}{2} = v_n + a_n dt + \frac{(a_{n+1} - a_n)dt}{2} = v_n + \frac{(a_{n+1} + a_n)dt}{2}$$

As a_{n+1} involves v_{n+1} for the drag force, we need to rearrange the above equation to get an expression for v_{n+1} in terms of known quantities only:

$$v_{n+1} = v_n + \frac{[F_r(n) + F_r(n+1) - \zeta(v_{n+1} + v_n) - k(x_{n+1} + x_n)]dt}{2m}$$

$$\Rightarrow v_{n+1} = \frac{1}{(1 + \zeta dt/2m)} \left(v_n + \frac{[F_r(n) + F_r(n+1) - \zeta v_n - k(x_{n+1} + x_n)]dt}{2m} \right)$$

4.2 Simulation I: A polystyrene bead in water

Now that we have the method ready, we try to simulate a realistic sample, a 1 μm diameter polystyrene bead in water at room temperature, $T = 295$ K. The mass of the bead is taken 0.55 pg and the viscosity of water is taken to be 890 $\mu\text{Pa}\cdot\text{s}$. Stokes’ law then gives the drag coefficient to be $\zeta = 6\pi(890)(0.5) \text{ pg/ms} \approx 8388 \text{ pg/ms}$. The slight discrepancy between the chosen values of viscosity and temperature does not affect our explorations ahead in any significant way.

We take the base units for mass, length and time as pg, μm and ms for our simulation. The units of force and energy become $\text{pg}\cdot\mu\text{m}\cdot\text{ms}^{-2} = \text{fN}$ (10^{-15}N , i.e. femto-Newton) and $\text{fN}\cdot\mu\text{m} = \text{zJ}$ (10^{-21}J , i.e. zepto-Joule).

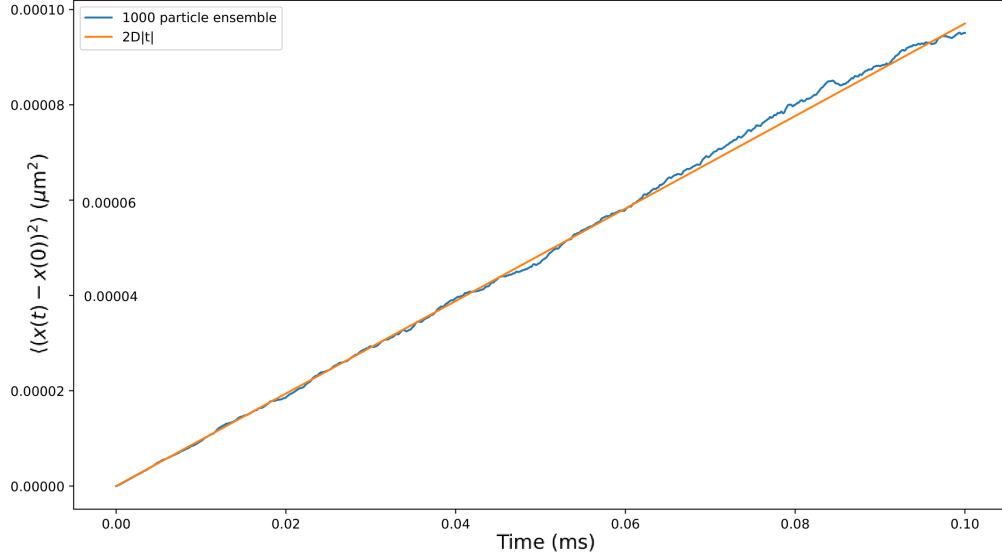
For our harmonic potential, we choose a trap stiffness of $k = 100 \text{ fN}/\mu\text{m}$, reasonable for an experimental optical trap. This gives us the values of velocity relaxation time and position relaxation time τ_v and τ as:

$$\tau_v = \frac{m}{\zeta} = \frac{0.55}{8388} = 6.55 \times 10^{-5} \text{ ms}, \quad \tau = \frac{\zeta}{k} = 83.88 \text{ ms}$$

We now attempt to verify some of the obtained correlation functions via our underdamped Langevin simulation method.

Verifying the Einstein relation

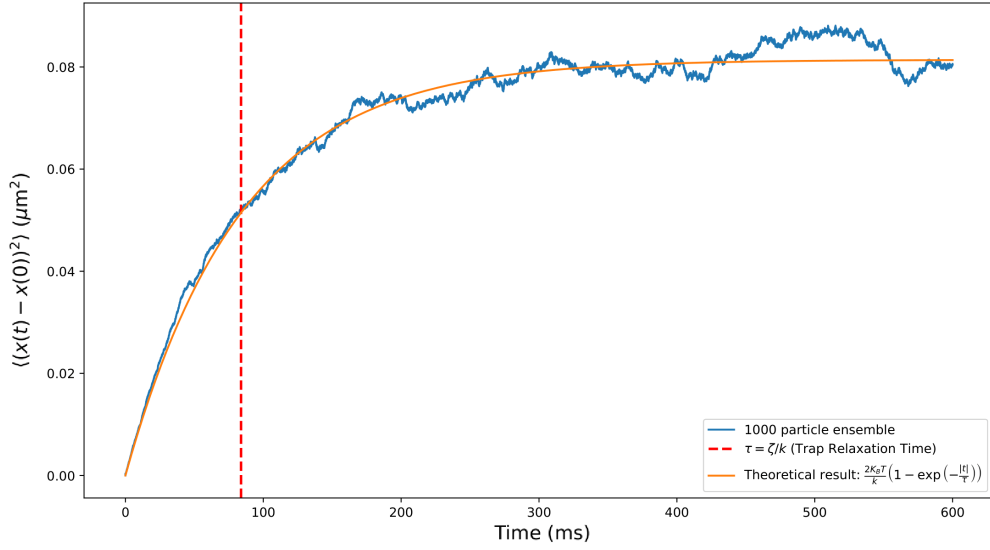
To verify the Einstein relation, we take ensemble average over 1000 simulated particles. We initialize the position and velocities of the particles by picking 1000 samples from the corresponding Boltzmann distributions $\mathcal{N}(0, \sqrt{k_B T/k})$ and $\mathcal{N}(0, \sqrt{k_B T/m})$. Then, we run our algorithm on each particle with $dt = 10^{-6}\text{ms}$ and 10^5 timesteps, corresponding to a total runtime of $dt \times \text{timesteps} = 0.1\text{ms}$. We then plot the mean square displacement over time, found by subtracting initial positions from current positions, squaring and then taking mean over all particles.



As $\tau_v \ll 0.1\text{ms}$, we see the characteristic diffusive growth of mean square displacement with slope $2D$ with $D = k_B T / \zeta$ as given by the Einstein relation. Also, as 0.1ms lies in the $\tau_v \ll t \ll \tau$ regime, this linear trend persists even with the harmonic potential in place, as derived earlier. As the slope is independent of trap stiffness k , we do not need to verify diffusion separately for the free particle where $k = 0$.

Relaxation of diffusion in a trap

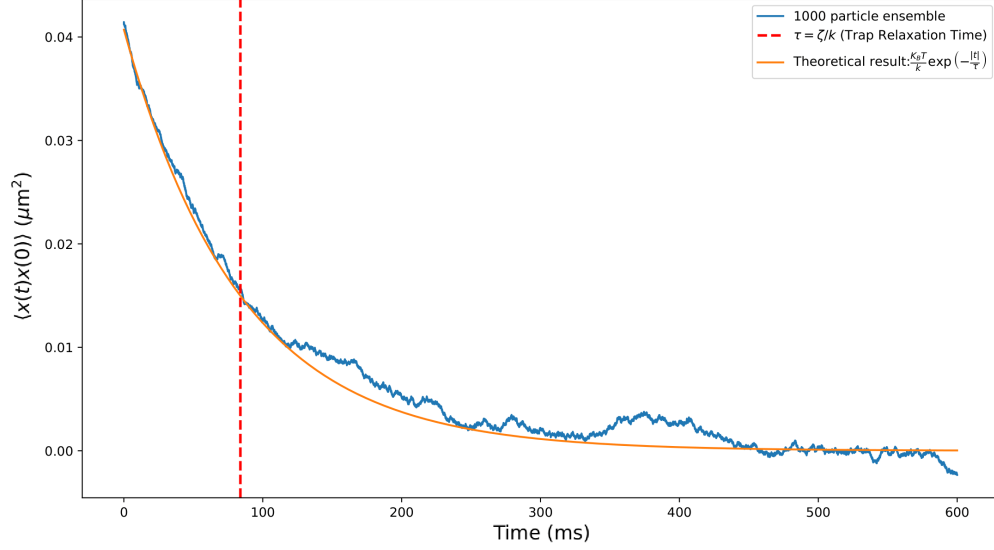
We now repeat the same simulation but with $dt = 10^{-3}\text{ms}$ and 6×10^5 timesteps, corresponding to total runtime 600 ms.



We observe a verification of the exponential saturation of mean square deviation due to confinement by the harmonic potential (trap) instead of continued linear diffusion curve as for a free particle. The fluctuating deviations from theoretical curve $(2k_B T/k)(1 - \exp(-t/\tau))$ are a consequence of 1000 being a relatively small number of particles in an ensemble, leading to incomplete cancellation of the deviations on averaging.

Position autocorrelation in a harmonic potential

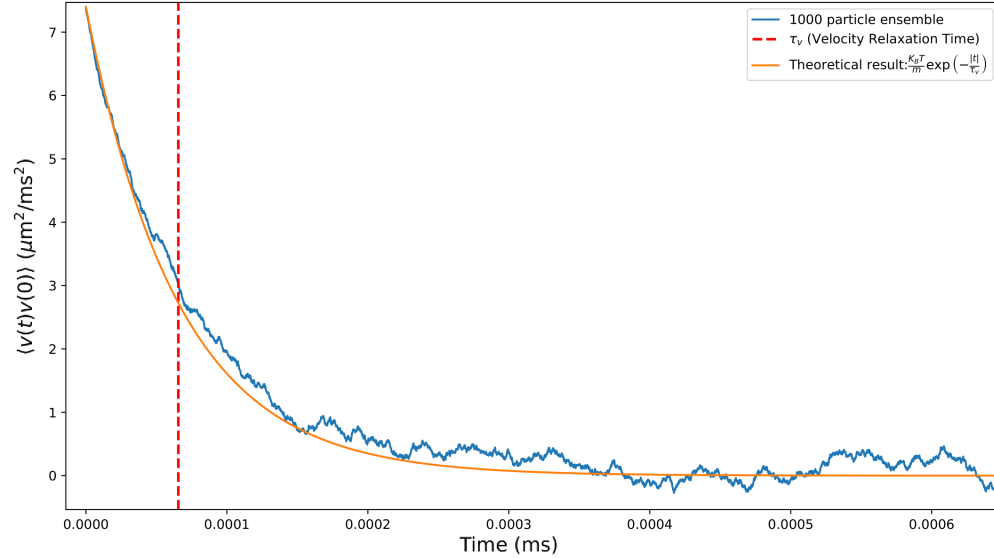
Now, we plot the position autocorrelation function for the harmonic potential, found by the multiplying positions for each particle with corresponding initial position and taking the mean over all particles. Again, with $dt = 10^{-3}\text{ms}$ and 6×10^5 timesteps, we run the simulation for 600ms:



Again, we see the simulated trajectories follow the theoretical curve $k_B T/k \exp(-|t|/\tau)$ closely, showing fluctuations due to the small sample size of 1000 particles. We recall that in presence of a potential, although we took the overdamped approximation ($\tau_v \rightarrow 0$) to derive our theoretical correlation functions, our underdamped simulation method agrees with it for times $t \gg \tau_v$, thus validating the approximation.

Relaxation of velocity autocorrelation

Now to study velocity relaxation, we plot the velocity autocorrelation analogously to the position autocorrelation. As τ_v is very small, we take $dt = 5 \times 10^{-8}$ ms and 14000 timesteps corresponding to a run time of 0.0007 ms, which is on the order of $10\tau_v$ and so is much less than τ . From our prior discussions we expect the theoretical curve in the harmonic trap for this regime ($t \ll \tau$) to act like a free particle.



We observe excellent agreement with the characteristic free particle decay curve $(k_B T/m) \exp(-|t|/\tau_v)$. As $0.0007 \text{ ms} \ll \tau$, we see this free particle behavior despite the harmonic potential which matches our earlier discussions. This simulation hence resolves that in this regime $\langle v(t)v(0) \rangle = 2D\delta(t)$ to be more precisely $(k_B T/m) \exp(-|t|/\tau_v)$, a piece of information lost due in the overdamped ($\tau_v \rightarrow 0$) limit.

When the time elapsed is on the order of τ , we write $\langle v(t)v(0) \rangle = D[2\delta(t) - \exp(-|t|/\tau)]$; recognizing the first term as the free particle velocity autocorrelation, the ratio of the coefficients of exponentials of second term to first is given by:

$$\frac{D/\tau}{K_B T/m} = \frac{\tau_v}{\tau} = \frac{6.55 \times 10^{-5}}{83.88} = 7.8 \times 10^{-7}$$

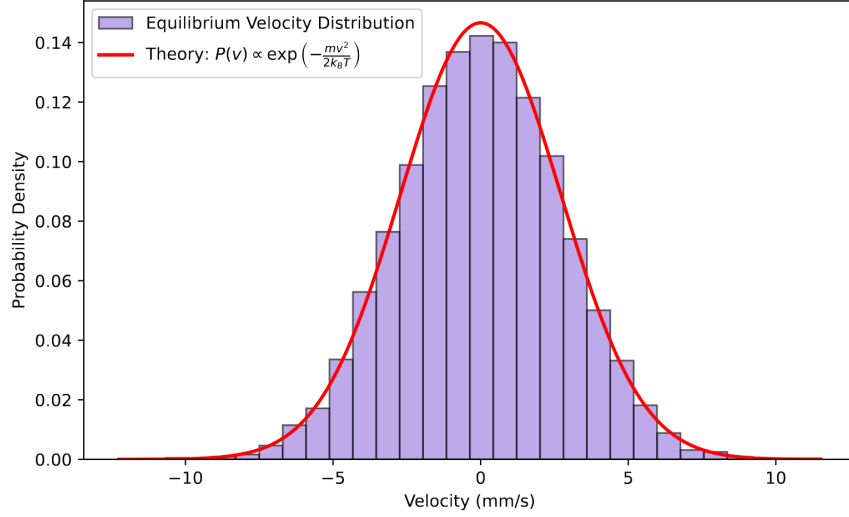
Thus, the second term is negligible with the parameters we have chosen and so will remain effectively invisible in our simulation even for longer times. We can thus expect the effect of a confining potential to be negligible for most real Brownian particles where $\tau_v \ll \tau$.

4.3 Simulation II: Boltzmann equilibration

To justify sampling the particles from the Boltzmann distribution in the previous simulation, we aim to show that the same dynamics actually lead to the equilibrium Boltzmann distributions.

Velocity equilibration

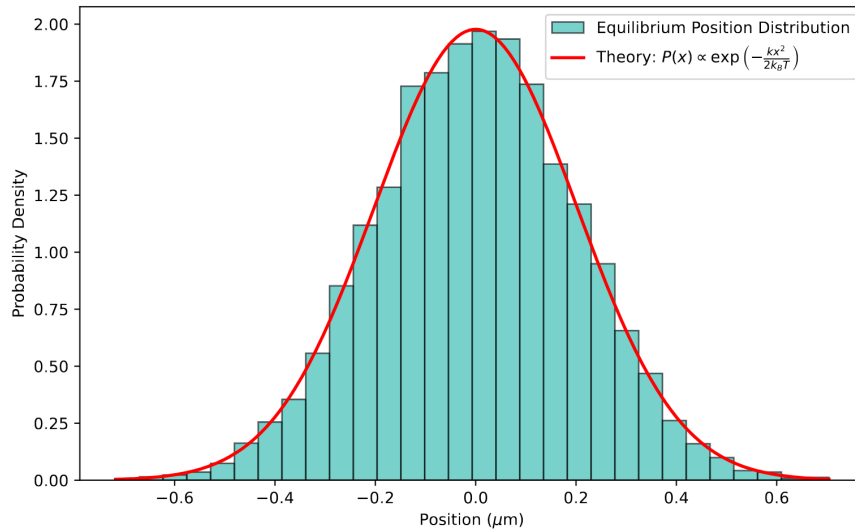
To verify velocity equilibration we choose $dt = 10^{-8}$ ms $\ll \tau_v$ and run the simulation for 5×10^4 steps (total runtime = 5×10^{-4} ms) starting a particle at rest at the minimum of the harmonic potential. To build a set of velocity measurements, we record the velocity at the last step and repeat it to get 10^4 data points and plot a density histogram as follows:



Thus, we see that the velocity histogram fits nicely into the Boltzmann envelope.

Position equilibration

For position equilibration, we have to choose $dt \gg \tau_v$ as $\tau \gg \tau_v$ makes $dt \ll \tau_v$ require a computationally unfeasible number of steps to reach τ timescales. Thus, we take $dt = 2 \times 10^{-3}$ ms for 8×10^4 steps (total runtime = 160 ms) and make a similar histogram with 10^5 data points:



Again, we see good agreement with the Boltzmann distribution.

Simultaneous equilibration

To fully justify their use in the previous simulation, we have to ensure the equilibrium in position and velocity both happen simultaneously. However, when we see the velocity distribution in the above $dt \gg \tau_v$ run, it has a variance much lower than the theoretical Boltzmann distribution while still retaining the Gaussian shape. This discrepancy is resolved by recognizing that when $dt \gg \tau_v$, the velocities in the simulation no longer represent instantaneous physical velocities. We now try to derive an expression for the variance of this simulated velocity. First, we rewrite the velocity update formula as:

$$v_{n+1} = av_n + b[F_r(n) + F_r(n+1)] - \frac{\zeta}{2m(1+\alpha)} \cdot \frac{dt}{\tau}$$

where:

$$\alpha = \frac{\zeta dt}{2m} = \frac{dt}{2\tau_v}, \quad a = \frac{(1-\alpha)}{(1+\alpha)} \quad \text{and} \quad b = \frac{\alpha}{\zeta(1+\alpha)}$$

As $dt \ll \tau$ in our simulations, we ignore the last term as an approximation (to avoid dealing with position and velocity cross-correlations). Now, taking average on both sides, using stationarity $\langle v_n \rangle = \langle v_{n+1} \rangle = \mu$ and $\langle F_r(n) \rangle = 0$, we get:

$$\mu = a\mu \implies \mu = 0 \quad (\because a \neq 1)$$

Thus, the variance is given by $\sigma^2 = \langle v_n^2 \rangle = \langle v_{n+1}^2 \rangle$ (again by stationarity at equilibrium). To evaluate it, we use the correlations:

$$\langle v_n F_r(n+1) \rangle = 0, \quad \langle F_r(n) F_r(m) \rangle = 2\delta_{mn} \zeta k_B T / dt = c\delta_{mn} \text{ (say)} \quad \text{and} \quad \langle v_n F_r(n) \rangle = \langle v_{n+1} F_r(n+1) \rangle = bc$$

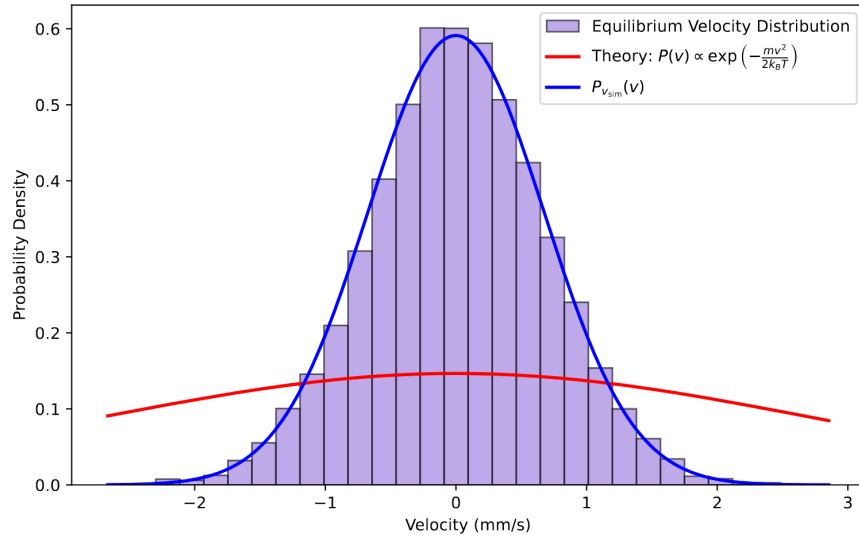
which give:

$$\begin{aligned} \langle v_{n+1}^2 \rangle &= a^2 \langle v_n^2 \rangle + b^2 (\langle F_r(n)^2 \rangle + \langle F_r(n+1)^2 \rangle) + 2ab \langle v_n F_r(n) \rangle \\ \implies \sigma^2 &= a^2 \sigma^2 + 2b^2 c + 2ab^2 c \Rightarrow \sigma^2 = \frac{2b^2 c}{1-a} \end{aligned}$$

Simplifying, we get:

$$\sigma^2 = \frac{2}{1 - \frac{(1-\alpha)}{(1+\alpha)}} \cdot \frac{\alpha^2}{\zeta^2(1+\alpha)^2} \cdot \frac{2\zeta k_B T}{dt} = \frac{4k_B T}{\zeta dt} \cdot \frac{\alpha^2}{2\alpha(1+\alpha)} = \frac{2k_B T}{\zeta dt} \cdot \frac{\zeta dt}{2m} \cdot \frac{1}{(1+\alpha)} = \frac{k_B T}{m(1+\alpha)} = \boxed{\frac{k_B T}{m(1+dt/2\tau_v)}}$$

Thus, we plot a Gaussian distribution $P_{v_{\text{sim}}} = \mathcal{N}(0, \sigma^2)$ alongside our ‘problematic’ velocity histogram:



Thus, the histogram indeed fits neatly under the $P_{v_{\text{sim}}}$ curve. In the $dt \ll \tau_v$ limit, $\sigma^2 = k_B T / m$ retrieves the Boltzmann distribution (as confirmed by earlier run). So, even while showing simultaneous Boltzmann equilibrium with $dt \ll \tau_v$ remains unfeasible, as the position distribution is independent of dt , we can say with confidence it is attained together with the velocity Boltzmann distribution eventually (as a smaller dt corresponds to a more accurate simulation of Langevin dynamics). This also means that in Simulation I, we should have sampled velocities from $P_{v_{\text{sim}}}$ for the runs with $dt \gg \tau_v$. However, running a simulation initialized with the Boltzmann distribution shows the velocity histogram relaxes to $P_{v_{\text{sim}}}$ in ~ 20 steps, which is a negligible fraction of the total number of steps, making a repetition of Simulation I with this corrected distribution unnecessary for our purposes.

5 Appendix

5.1 The Probability Current and Detailed Balance

In this section, we aim to show that the apparent freedom to add a function of time to J in the Fokker-Planck continuity equation can be done away by deriving the expression for J independently via consideration of probabilities of the motion of particles. Let $w(x, \Delta x; \Delta t)$ denote the probability density that a particle at x moves to $x + \Delta x$ in time Δt at a fixed time t . Thus, the probability that a particle at x at time t will move to $x + \Delta x$ in time Δt per unit time is $P(x, t)w(x, \Delta x; \Delta t)/\Delta t$. In the continuum limit, we take $\Delta t \rightarrow 0$ and formulate J at position x_0 and time t as the sum of all probability flows passing through the point x_0 . The probability flow per unit time passing through x_0 from the left to right is then given by:

$$J_{\text{right}} = \lim_{\Delta t \rightarrow 0} \left[\frac{1}{\Delta t} \int_0^\infty d\Delta x \int_{x_0 - \Delta x}^{x_0} P(x, t) w(x, \Delta x; \Delta t) dx \right]$$

where the limits are chosen for so that x lies to the left of x_0 and $x + \Delta x$ lies to the right. Similarly, for accounting for flow from right to left:

$$J_{\text{left}} = \lim_{\Delta t \rightarrow 0} \left[\frac{1}{\Delta t} \int_{-\infty}^0 d\Delta x \int_{x_0}^{x_0 - \Delta x} P(x, t) w(x, \Delta x; \Delta t) dx \right]$$

Thus, the net current is given by (where the convention that $J > 0$ implies flow to the right is embedded in the continuity equation):

$$J = J_{\text{right}} - J_{\text{left}}$$

Taking $f(x) = P(x, t)w(x, \Delta x; \Delta t)$, we use the Taylor series of f about $x = x_0$ to evaluate the integrals:

- For J_{right} :

$$\begin{aligned} \int_{x_0 - \Delta x}^{x_0} P(x, t) W(x, \Delta x; \Delta t) dx &= \int_{x_0 - \Delta x}^{x_0} \left[\sum_{j=0}^{\infty} \frac{(x - x_0)^j}{j!} f^{(j)}(x_0) \right] dx = - \sum_{j=0}^{\infty} \frac{(-\Delta x)^{j+1}}{(j+1)!} f^{(j)}(x_0) \\ \Rightarrow J_{\text{right}} &= \lim_{\Delta t \rightarrow 0} \left[\frac{-1}{\Delta t} \sum_{j=0}^{\infty} \int_{-\infty}^0 d\Delta x \frac{(-\Delta x)^{j+1}}{(j+1)!} f^{(j)}(x_0) \right] \end{aligned}$$

- For J_{left} :

$$\begin{aligned} \int_{x_0}^{x_0 + \Delta x} P(x, t) W(x, \Delta x; \Delta t) dx &= \int_{x_0}^{x_0 + \Delta x} \left[\sum_{j=0}^{\infty} \frac{(x - x_0)^j}{j!} f^{(j)}(x_0) \right] dx = \sum_{j=0}^{\infty} \frac{(\Delta x)^{j+1}}{(j+1)!} f^{(j)}(x_0) \\ \Rightarrow J_{\text{left}} &= \lim_{\Delta t \rightarrow 0} \left[\frac{1}{\Delta t} \sum_{j=0}^{\infty} \int_0^\infty d\Delta x \frac{(\Delta x)^{j+1}}{(j+1)!} f^{(j)}(x_0) \right] \end{aligned}$$

Hence, on considering the net current, the integration limits combine to give:

$$J = \lim_{\Delta t \rightarrow 0} \left[\frac{1}{\Delta t} \sum_{j=0}^{\infty} \frac{(-1)^j}{(j+1)!} \int_{-\infty}^{\infty} (\Delta x)^{j+1} f^{(j)}(x_0) d\Delta x \right] = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \sum_{j=0}^{\infty} \frac{(-1)^j}{(j+1)!} \frac{\partial^j}{\partial x^j} \left(P(x, t) \int_{-\infty}^{\infty} (\Delta x)^{j+1} w(x, \Delta x; t) d\Delta x \right) \Big|_{x=x_0}$$

Now, we note that the term

$$\int_{-\infty}^{\infty} (\Delta x)^{j+1} w(x, \Delta x; t) d\Delta x$$

represents the $(j+1)^{\text{th}}$ raw moment of the random variable $\Delta X = \mu \Delta t + \sigma \Delta W$ as the process we are interested in is the Itô process. We know in the continuum limit ($\Delta t \rightarrow 0$):

$$\frac{\langle \Delta X \rangle}{\Delta t} = \mu(x, t) \quad \text{and} \quad \frac{\langle (\Delta X)^2 \rangle}{\Delta t} = \sigma^2(x, t)$$

using $\langle \Delta W \rangle = 0$ and $\langle (\Delta W)^2 \rangle = \Delta t$. However, for higher order ($j > 1$) terms, the order of $\langle (\Delta X)^{j+1} \rangle$ exceeds Δt and division by Δt combined with the $\Delta t \rightarrow 0$ limit forces these terms to vanish; Since μ and σ are assumed to be smooth, the

derivatives only affect the coefficients in the expansion and the order of Δt in the terms remains unchanged. Therefore, the final expression for J becomes:

$$J(x_0, t) = \mu(x_0, t)P(x_0, t) - \frac{1}{2} \frac{\partial(\sigma^2 P)}{\partial x} \Big|_{x=x_0}$$

which is exactly what we obtained using the Fokker-Planck equation, only this time without any freedom to add an arbitrary function of time. Now that we have connected the Fokker-Planck current to transition probabilities, we aim to justify why this distinguished J must vanish at equilibrium. Imagine a movie showing a system at equilibrium with microscopic resolution; By the time-reversal symmetry of physical laws (the Hamiltonian), the statistical behavior of particles should not change even when the movie is played in reverse, this condition is called **detailed balance**. In precise terms, this implies that the probability of a reverse process is same as the forward process at equilibrium (in the same time interval Δt), that is:

$$P_{\text{eq}}(x)w(x, \Delta x; \Delta t) = P_{\text{eq}}(x + \Delta x)w(x + \Delta x, -\Delta x; \Delta t)$$

To apply this criterion on J , we make transformations $\Delta x \rightarrow -\Delta x$ and $x + \Delta x \rightarrow x$ to make the limits of J_{left} same as J_{right} :

$$J_{\text{left}} = \lim_{\Delta t \rightarrow 0} \left[\frac{1}{\Delta t} \int_0^\infty d\Delta x \int_{x_0}^{x_0 + \Delta x} P(x, t)w(x, -\Delta x; \Delta t) dx \right] = \lim_{\Delta t \rightarrow 0} \left[\frac{1}{\Delta t} \int_0^\infty d\Delta x \int_{x_0 - \Delta x}^{x_0} P(x + \Delta x, t)w(x + \Delta x, -\Delta x; \Delta t) dx \right]$$

Thus, $J = J_{\text{right}} - J_{\text{left}}$ at equilibrium gives:

$$J = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int_0^\infty d\Delta x \int_{x_0 - \Delta x}^{x_0} [P_{\text{eq}}(x)w(x, \Delta x; \Delta t) - P_{\text{eq}}(x + \Delta x)w(x + \Delta x, -\Delta x; \Delta t)] dx$$

Therefore, the detailed balance criterion trivially forces the integrand to vanish, giving $J = 0$ at equilibrium justifying our assumption in the main text.

5.2 Brownian Dynamics in Generalized Coordinates

In the main text, we discussed Brownian motion in one dimension. However, particles in most soft matter systems possess both translational and rotational degrees of freedom. In this section, we briefly extend the overdamped Langevin equation to generalized coordinates and see how time-reversal symmetry once again yields valuable physical insight. For a particle of general shape, we introduce generalized coordinates:

$$x = (x_1, x_2, \dots, x_f)$$

where some x_i denote position or orientation coordinates. The potential force is given by

$$dU = - \sum_i F_{pi} dx_i$$

so that

$$F_{pi} = - \frac{\partial U}{\partial x_i}$$

The irreversible work per unit time is

$$W_{\text{irr}} = \sum_i F_{fi} \dot{x}_i$$

where the friction force is written as

$$F_{fi} = - \sum_j \zeta_{ij}(x) \dot{x}_j$$

Hence, the Langevin equation for generalized coordinates is

$$- \sum_j \zeta_{ij}(x) \dot{x}_j - \frac{\partial U}{\partial x_i} + F_{ri}(t) = 0$$

This comes from Newton's second law of motion $F_{pi} + F_{fi} + F_{ri} = \sum_j M_{ij} \ddot{x}_j$ where M_{ij} is the generalized inertia matrix. For a Cartesian position coordinate x_i , $M_{ij} = m\delta_{ij}$ whereas if x_i is an angular coordinate, M_{ij} is the inertia tensor. As this level of rigor is beyond our present scope, we simply note that the inertial contribution represented by M_{ij} is negligible on soft matter timescales. Thus frictional forces dissipate any acceleration quickly, and the acceleration terms may again be ignored, giving $F_{pi} + F_{fi} + F_{ri} = 0$ as expanded above.

The random force is assumed to satisfy

$$\langle F_{ri}(t) \rangle = 0$$

It also satisfies (to be derived in next section):

$$\langle F_{ri}(t) F_{rj}(t') \rangle = 2k_B T \zeta_{ij} \delta(t - t')$$

The friction matrix ζ_{ij} satisfies the **reciprocal relation**

$$\zeta_{ij}(x) = \zeta_{ji}(x)$$

and is positive definite:

$$\sum_{ij} \zeta_{ij}(x) \dot{x}_i \dot{x}_j \geq 0$$

Onsager's Proof of the Reciprocal Relation

One can prove the reciprocal relation via hydrodynamics, however, Onsager gave a proof only using time reversal symmetry of correlation function at equilibrium as presented ahead. Let:

$$\xi_i = x_i - x_i^0$$

denote the deviation from equilibrium (x_i^0 's correspond to equilibrium values). We examine the behavior under a harmonic potential:

$$U = \frac{1}{2} \sum_i k_i \xi_i^2$$

Then the Langevin equation becomes

$$-\sum_j \zeta_{ij} \dot{\xi}_j - k_i \xi_i + F_{ri}(t) = 0.$$

In matrix form, let

$$\boldsymbol{\xi} = (\xi_1, \xi_2, \dots, \xi_f)^T, \quad \mathbf{K} = \text{diag}(k_1, k_2, \dots, k_f), \quad \mathbf{F}_r = (F_{r1}, F_{r2}, \dots, F_{rf})^T.$$

Then

$$\zeta \dot{\boldsymbol{\xi}} + \mathbf{K} \boldsymbol{\xi} = \mathbf{F}_r(t).$$

So,

$$\dot{\boldsymbol{\xi}} = \zeta^{-1} (\mathbf{F}_r(t) - \mathbf{K} \boldsymbol{\xi}).$$

Let

$$\boldsymbol{\xi}(0) = \boldsymbol{\xi}_0.$$

For a small time interval Δt (integrating both sides from 0 to Δt),

$$\boldsymbol{\xi}(\Delta t) = \boldsymbol{\xi}_0 - \Delta t \zeta^{-1} \mathbf{K} \boldsymbol{\xi}_0 + \int_0^{\Delta t} dt_1 \zeta^{-1} \mathbf{F}_r(t_1).$$

In components,

$$\xi_i(\Delta t) = \xi_{i0} - \Delta t \sum_k (\zeta^{-1})_{ik} k_k \xi_{k0} + \sum_k \int_0^{\Delta t} dt_1 (\zeta^{-1})_{ik} F_{rk}(t_1).$$

Now multiply by ξ_{j0} and take the ensemble average:

$$\langle \xi_i(\Delta t) \xi_{j0} \rangle = \langle \xi_{i0} \xi_{j0} \rangle - \Delta t \sum_k (\zeta^{-1})_{ik} k_k \langle \xi_{k0} \xi_{j0} \rangle.$$

Here, the integral term drops out because $\langle F_{rk}(t_1) \rangle = 0$. At equilibrium, by the Boltzmann distribution, we have:

$$\langle \xi_{i0} \xi_{j0} \rangle = \delta_{ij} \frac{k_B T}{k_i}.$$

Therefore,

$$\frac{1}{\Delta t k_B T} \left[\delta_{ij} \frac{k_B T}{k_i} - \langle \xi_i(\Delta t) \xi_{j0} \rangle \right] = (\zeta^{-1})_{ij}$$

By time-reversal symmetry at equilibrium, we have the $i - j$ symmetry discussed earlier:

$$\langle \xi_i(\Delta t) \xi_{j0} \rangle = \langle \xi_j(\Delta t) \xi_{i0} \rangle,$$

Also, as $\delta_{ij} = \delta_{ji}$ the left-hand side is completely symmetric with respect to swapping i and j , thus the right-hand side must also be symmetric in this way:

$$(\zeta^{-1})_{ij} = (\zeta^{-1})_{ji},$$

This gives the desired reciprocal relation:

$$\boxed{\zeta_{ij} = \zeta_{ji}}$$

Finally, assuming the random force correlation takes the white-noise form:

$$\langle F_{ri}(t) F_{rj}(t') \rangle = A_{ij} \delta(t - t').$$

We return to the earlier expression:

$$\xi_i(\Delta t) = \xi_{i0} - \Delta t \sum_k (\zeta^{-1})_{ik} k_k \xi_{k0} + \sum_k \int_0^{\Delta t} dt_1 (\zeta^{-1})_{ik} F_{rk}(t_1).$$

Multiplying by $\xi_j(\Delta t)$ and taking the ensemble average gives

$$\begin{aligned} \langle \xi_i(\Delta t) \xi_j(\Delta t) \rangle &= \left\langle \left[\xi_{i0} - \Delta t \sum_k (\zeta^{-1})_{ik} k_k \xi_{k0} + \sum_k \int_0^{\Delta t} dt_1 (\zeta^{-1})_{ik} F_{rk}(t_1) \right] \right. \\ &\quad \times \left. \left[\xi_{j0} - \Delta t \sum_{k'} (\zeta^{-1})_{jk'} k_{k'} \xi_{k'0} + \sum_{k'} \int_0^{\Delta t} dt_2 (\zeta^{-1})_{jk'} F_{rk'}(t_2) \right] \right\rangle \end{aligned}$$

As Δt is small, we keep terms only up to first order in Δt to get (other terms are omitted as $\langle F_{rk} \rangle = 0$):

$$\begin{aligned} \langle \xi_i(\Delta t) \xi_j(\Delta t) \rangle &= \langle \xi_{i0} \xi_{j0} \rangle - \Delta t \sum_k (\zeta^{-1})_{ik} k_k \langle \xi_{k0} \xi_{j0} \rangle \\ &\quad - \Delta t \sum_{k'} (\zeta^{-1})_{jk'} k_{k'} \langle \xi_{i0} \xi_{k'0} \rangle + \sum_{k,k'} (\zeta^{-1})_{ik} (\zeta^{-1})_{jk'} \\ &\quad \times \int_0^{\Delta t} dt_1 \int_0^{\Delta t} dt_2 \langle F_{rk}(t_1) F_{rk'}(t_2) \rangle. \end{aligned}$$

Using $\langle F_{rk}(t_1) F_{rk'}(t_2) \rangle = A_{kk'} \delta(t_1 - t_2)$, the double integral becomes :

$$\int_0^{\Delta t} dt_1 \int_0^{\Delta t} dt_2 A_{kk'} \delta(t_1 - t_2) = A_{kk'} \Delta t.$$

Hence,

$$\begin{aligned} \langle \xi_i(\Delta t) \xi_j(\Delta t) \rangle &= \langle \xi_{i0} \xi_{j0} \rangle - \Delta t \sum_k (\zeta^{-1})_{ik} k_k \langle \xi_{k0} \xi_{j0} \rangle \\ &\quad - \Delta t \sum_{k'} (\zeta^{-1})_{jk'} k_{k'} \langle \xi_{i0} \xi_{k'0} \rangle + \Delta t \sum_{k,k'} (\zeta^{-1})_{ik} (\zeta^{-1})_{jk'} A_{kk'}. \end{aligned}$$

Using the earlier result $\langle \xi_{i0} \xi_{j0} \rangle = \delta_{ij} k_B T / k_i$, we get:

$$\begin{aligned} \langle \xi_i(\Delta t) \xi_j(\Delta t) \rangle &= \delta_{ij} \frac{k_B T}{k_i} - 2 \Delta t k_B T (\zeta^{-1})_{ij} \\ &\quad + \Delta t \sum_{k,k'} (\zeta^{-1})_{ik} (\zeta^{-1})_{jk'} A_{kk'}. \end{aligned}$$

By time-translation symmetry in equilibrium we also know:

$$\langle \xi_i(\Delta t) \xi_j(\Delta t) \rangle = \langle \xi_{i0} \xi_{j0} \rangle = \delta_{ij} k_B T / k_i$$

Thus,

$$2 k_B T (\zeta^{-1})_{ij} = \sum_{k,k'} (\zeta^{-1})_{ik} (\zeta^{-1})_{jk'} A_{kk'}$$

Defining the matrix $(A)_{ij} = A_{ij}$, we write the above equation in matrix form. Since $\sum_{k,k'} (\zeta^{-1})_{ik} (\zeta^{-1})_{jk'} A_{kk'}$ is precisely the (i, j) -component of $\zeta^{-1} A (\zeta^{-1})^T$, we get:

$$2k_B T \zeta^{-1} = \zeta^{-1} A (\zeta^{-1})^T$$

Using the reciprocal relation,

$$(\zeta^{-1})^T = \zeta^{-1},$$

This gives:

$$2k_B T \zeta^{-1} = \zeta^{-1} A \zeta^{-1}$$

Multiplying from the left by ζ ,

$$2k_B T I = A \zeta^{-1}.$$

Multiplying from the right by ζ ,

$$2k_B T \zeta = A$$

Hence, in component form,

$$\boxed{A_{ij} = 2k_B T \zeta_{ij}}$$

Therefore,

$$\boxed{\langle F_{ri}(t) F_{rj}(t') \rangle = 2k_B T \zeta_{ij} \delta(t - t')}$$

This is the second fluctuation-dissipation relation for generalized coordinates.